7. Suppose you have a rose plant growing in a pot, how will you demonstrate transpiration in it?
8. Mention the differences between stomatal and lenticular transpiration
9. To which directional stimuli do (a) roots respond (b) shoots respond?

VII. Answer in detail.

1. Differentiate between tropic and nastic movements
2. How will you differentiate the different types of transpiration?

VIII. Higher Order Thinking Skills.

1. There are 3 plants A, B and C. The flowers of A open their petals in bright light during the day but closes when it gets dark at night. On the other hand, the flowers of plant B open their petals at night but closes during the day when there is bright light. The leaves of plant C fold up and droop when touched with fingers or any other solid object.
   a) Name the phenomenon shown by the flowers of plant A and B.
   b) Name one plant each which behaves like the flowers of plant A and B.
   c) Name the phenomenon exhibited by the leaves of plant C.
   d) Name the plant which behaves like the leaves of plant 'C'?
2. Imagine that student A studied the importance of certain factors in photosynthesis. He took a potted plant and kept it in dark for 24 hours. In the early hours of the next morning, he covered one of the leaves with dark paper in the centre only. Then he placed the plant in sunlight for a few hours and tested the leaf which was covered with black paper for starch.
   a) What aspect of photosynthesis was being investigated?
   b) Why was the plant kept in the dark before the experiment?
   c) How will you prove that starch is present in the leaves?
   d) Name the raw materials needed for photosynthesis.

REFERENCE BOOKS

1. Devlin and Witham, 1986. Plant Physiology, 1st edition
2. B.P. Pandey 2003 Modern Practical Botany Vol. II.
3. V.K. Jain 2003 Plant physiology.

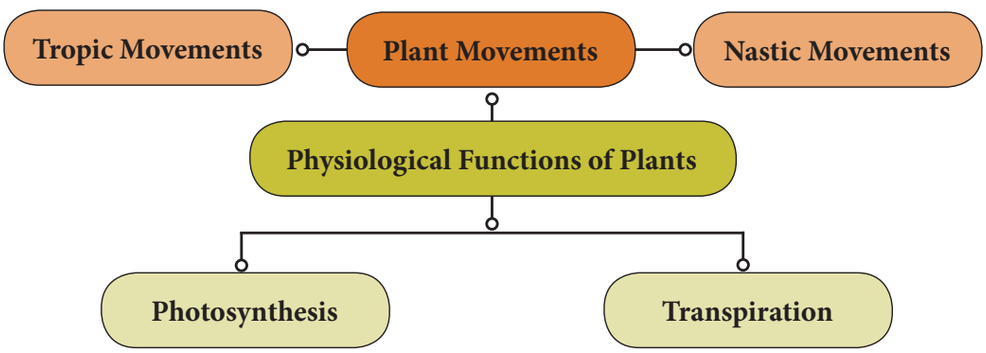
INTERNET RESOURCES

http://web.mit.edu/esgbio

http://www.bioedonline.org/

http://www.biology.arizona.edu/default.html

Concept Map



IX_SCI_EM_Unit-19_BIO.indd 232 06-02-2020 13:14:56